CareerPilot

CAREERPILOT NIT3004 PROGRESS DIARY

Smart CV application powered by Agentic AI

Abstract

An application that helps you navigate into your desired career

[Author name]

[Email address]

Introduction

CareerPilot is a multi-agent AI platform that improves job seekers' employability. A user uploads a CV and receives matched jobs, a skill gap analysis, CV feedback, a study plan, interview practice and a readiness report. This diary records the work done on each subsystem, with the files changed, the date, the team member responsible and the technical detail of each change.

Team members: Agrata Shrestha, Ujjwol Adhikari

Unit: NIT3004

Supervisor: Gongqi Lin

Repository: https://github.com/Ujjwol77/careerpilot

The platform has several agents run in sequence by an orchestrator: Profile Understanding, Job Search, Job Matching, Skill Gap, CV Optimisation, Learning Pathway and Interview Coach. A Report agent combines their outputs into a readiness score. If job search returns fewer than 3 postings, or the best match scores under 35, the orchestrator relaxes the filters and retries. This retry is the agentic reflection step.

Layer	Technology
Frontend	Next.js 16 (App Router), React 18.3, TypeScript 5.5
Backend API	Python, FastAPI, Uvicorn
Database	SQLAlchemy 2 ORM on SQLite (PostgreSQL/Supabase-ready via `DATABASE_URL`)
Authentication	PBKDF2 password hashing (200,000 rounds), signed JWT sessions (PyJWT)
CV parsing	pypdf, python-docx
Language model	Google Gemini through its OpenAI-compatible endpoint (httpx client)
Live job data	Adzuna Jobs API (Australia)
Configuration	`backend/.env` loaded with python-dotenv
Testing and CI	pytest, GitHub Actions workflow

Learning Pathway Subsystem

The Learning Pathway agent now links every skill gap to a specific course page instead of a Google search. This addressed feedback that learning materials were not specific enough.

Files used
`backend/app/data/learning.py`

Files used
`backend/app/agents/learning_agent.py`
`backend/app/check_links.py`
`backend/tests/test_api.py`
`frontend/app/dashboard/learning/page.tsx`
`README.md`

Date	Team member	Notes
05/11/2026	Agrata	**Feedback received:** learning items only produced a Google search query (`search` field), and the hours were placeholder values. Reviewed the existing catalogue: 23 skills had descriptive titles and every other skill fell back to a generic "Introductory course" search.
05/11/2026	Agrata	**Curated catalogue:** rewrote `learning.py` around a `Resource` dataclass (type, title, provider, url, hours, cost, hours_source). All 71 skills in the taxonomy now map to at least one direct page on a named provider: Kaggle Learn, Microsoft Learn, TryHackMe, Cisco Networking Academy, Professor Messer, PortSwigger, MDN, Harvard CS50, MIT OpenCourseWare, official framework docs and certification exam pages. Soft skills keep a practice activity plus a guide link (e.g. Toastmasters, Atlassian Team Playbook).
05/11/2026	Agrata	**Honest effort estimates:** where a provider publishes a duration (e.g. TryHackMe SOC Level 1 = 65 h, Microsoft Learn PL-300T00 = 72 h) it is used and tagged `hours_source: "provider"`; all other hours are tagged `"estimate"`. The UI prefixes estimates with "~".
05/11/2026	Agrata	**Link verification:** checked about 20 of the least certain URLs through web fetch. One had moved (AWS Cloud Practitioner page now at `/training/learn-about/cloud-practitioner/`) and was updated; the Australian Cyber Security Centre page blocks automated checks, so it was replaced with the NIST Cybersecurity Framework. Added `python -m app.check_links`, which sends a HEAD request (falling back to GET) to every catalogue URL and exits with code 1 if any link is broken, so it can run in CI.
05/11/2026	Agrata	**Agent change:** `build_plan()` now tracks seen URLs so a course that covers two gaps (e.g. Cisco Network Defense for

Date	Team member	Notes
		both network security and firewalls) is scheduled once, not twice.
05/11/2026	Agrata	**Frontend:** the learning page now renders each course title as a direct link with provider, cost, hours and scheduled weeks, replacing the "Find resources" Google link.
05/11/2026	Agrata	**Testing:** added `test_every_skill_has_direct_learning_links` (every skill has an https link that is not a search page) and `test_learning_plan_items_link_to_course_pages` (plan URLs are unique). An early version of the first test wrongly rejected Google's own Machine Learning Crash Course because it matched on the domain `google.`; the check was changed to reject search paths and `q=` query strings instead. Result: 28 of 28 backend tests passing.

Language Model (Gemini) Subsystem

The backend can now call Google Gemini for CV rewrites, interview feedback and the report summary, with the rule-based templates kept as a fallback. Gemini was chosen over a local model (Ollama) because it runs in the cloud, so teammates working remotely get the same model behaviour.

Files used
`backend/app/config.py`
`backend/app/llm/__init__.py`
`backend/app/check_llm.py`
`backend/app/prompt_bank.py`
`backend/requirements.txt`
`backend/.env.example`

Date	Team member	Notes
05/11/2026	Agrata	**Review of the LLM layer:** `LLM_PROVIDER=mock` (the default) returns hand-written template text and never calls a model. The `OpenAICompatLLM` class posts to `{base_url}/chat/completions`, but it had only been tested with `httpx.MockTransport` (a fake server), never against a real provider. Only three prompts use the model: `cv_rewrite.v1`, `interview_feedback.v1`,

Date	Team member	Notes
		`report_summary.v1`.
07/11/2026	Agrata	**Provider decision:** compared Ollama (local, free), Groq/Gemini (cloud, free tier) and OpenAI/Claude (paid). Chose Gemini through its OpenAI-compatible endpoint `https://generativelanguage.googleapis.com/v1beta/openai/`, so no new client code was needed. Model set to `gemini-3.5-flash-lite` (the stable `gemini-3.6-flash` is the alternative).
07/11/2026	Agrata	**Bug found and fixed:** `config.py` read settings only with `os.getenv()`, and nothing loaded `backend/.env`, so an API key placed in `.env` was silently ignored and the app stayed in mock mode. Added `python-dotenv` to `requirements.txt` and `load_dotenv(..., override=False)` at the top of `config.py`, so real environment variables on a host still take priority. Tests are unaffected because `conftest.py` sets `LLM_PROVIDER=mock` before import.
07/11/2026	Agrata	**Connection checker:** added `python -m app.check_llm`, which prints the provider, URL, model and a masked key, sends a one-line test prompt and maps failures to causes (400/404 = model or URL, 401/403 = key, 429 = quota). Tested with a dummy key: `.env` was read and the request reached Google (403 returned as expected).
07/11/2026	Agrata	**Setup issues resolved:** `pip` and `ModuleNotFoundError: dotenv` errors were caused by running commands outside the activated virtual environment (`.venv\Scripts\Activate.ps1`). Documented the activation step for teammates.
07/11/2026	Agrata	**Security:** `.env` is listed in `.gitignore`, so keys are never committed; each team member generates their own key in Google AI Studio. Contact details are already redacted (`redact_pii`) before any CV text is sent to the model.

Live Job Search (Adzuna API) Subsystem

The Job Search agent can now pull real Australian postings from the Adzuna API for the user's target role and location, and falls back to the 45 synthetic postings when no key is set or the API fails.

Files used
`backend/app/agents/job_search_agent.py`

Files used
`backend/app/agents/job_match_agent.py`
`backend/app/models.py`
`backend/app/main.py`
`backend/app/config.py`
`backend/app/routers/jobs.py`
`backend/app/check_jobs.py`
`backend/tests/test_api.py`, `backend/tests/conftest.py`
`backend/.env.example`, `README.md`
`frontend/app/dashboard/jobs/page.tsx`

Date	Team member	Notes
08/11/2026	Agrata	**API research:** compared free-tier job APIs. Adzuna: Australian coverage (`au`), free key issued instantly, returns title, company, location, description snippet, salary and apply link. JSearch (Google for Jobs data): widest coverage but only 200 requests per month on the free plan. Jooble: free key by request form. Careerjet: requires the end user's IP and user agent on every call. Arbeitnow/Remotive: Europe and remote only. SEEK, LinkedIn and Indeed have no public search API. Chose Adzuna as the primary source.
08/11/2026	Agrata	**Terms review:** Adzuna permits personal and academic research use with attribution ("The Adzuna API", linked); use by an organisation is limited to a 14-day trial without written consent. A "Jobs by Adzuna" credit was added to the Jobs page.
08/11/2026	Agrata	**`AdzunaSource` implementation:** a new class implementing the existing `JobSource` protocol. It calls `GET https://api.adzuna.com/v1/api/jobs/au/search/1` with `what` = the role label (e.g. "Data Analyst"), `where` = the preferred location, `results_per_page` = 30 and `max_days_old` = 30. Each result is cleaned of HTML tags, its skills are extracted with the existing taxonomy, an embedding is computed, and its level is inferred from the title by regex (senior/lead/principal → senior; junior/graduate/entry/intern → junior; otherwise mid, mapped to 0/2/5 minimum years). Rows are upserted on

Date	Team member	Notes
		(`source`, `external_id`) so a repeated posting updates rather than duplicates.
08/11/2026	Agrata	**Caching and fallback:** a new `job_query_cache` table stores the time of each search (key = country, role, location). Repeat searches within `JOBS_CACHE_HOURS` (default 12) are served from the database with no API call, protecting the free quota. `search_jobs()` queries live sources first; if they return fewer than 3 postings, the seed dataset is added. API errors are logged and saved results are returned.
08/11/2026	Agrata	**Database migration:** `Base.metadata.create_all()` does not alter existing tables, so existing team databases would fail on the new `url` and `external_id` columns. Added `add_missing_columns()` to the startup lifespan, which inspects the `jobs` table and runs `ALTER TABLE ... ADD COLUMN` only for missing columns. Verified on a database built with the old schema: columns were added and existing jobs still loaded.
08/11/2026	Agrata	**Filter bug fixed:** the minimum-salary filter removed every posting with no listed salary (stored as 0). Unknown salaries are now kept, and the frontend shows "Salary not listed" instead of "$0k–$0k".
08/11/2026	Agrata	**Frontend:** match results and search results show a "View and apply on Adzuna" link and a source note. `source` and `url` are now passed through `match_jobs()` and the `/jobs` endpoints.
08/11/2026	Agrata	**Testing:** three new tests use `httpx.MockTransport` as a fake Adzuna server: results are saved with correct level, skills and URL, and a second identical search makes no extra API call; a 401 error falls back to sample jobs; no key means sample jobs only. `conftest.py` blanks the Adzuna keys so a developer's `.env` never makes tests call the real API. Result: 31 of 31 tests passing; frontend type-check and production build clean. Added `python -m app.check_jobs "<role>" <location>` to test a real key.
08/11/2026	Agrata	**Known limitation:** Adzuna returns only a description snippet, so skills listed further down the full advert are missed and live match scores can be lower than for the sample jobs.

Profile Understanding Subsystem

The profile page now shows the roles that fit the CV as readable names with a fit score, instead of "[object Object]".

Files used
`frontend/app/dashboard/profile/page.tsx`
`backend/app/agents/profile_agent.py` (reviewed, not changed)
`backend/samples/sample_cv_cyber.pdf`

Date	Team member	Notes
05/11/2026	Agrata	**Second test CV:** created `sample_cv_cyber.pdf`, a one-page CV for a fictional junior cyber security analyst, with ReportLab. The skill extractor detects 21 skills from it. It deliberately omits SIEM, cloud and threat modelling so the skill gap and learning pages have gaps to show, and uses weak bullets ("Responsible for…", "Worked on…") to exercise the CV review agent. Known quirk: "Currently studying for CompTIA Security+" is still counted as having Security+.
08/11/2026	Agrata	**Bug:** the "roles that fit" field showed "[object Object], [object Object], [object Object]". `infer_roles()` returns objects `{role, label, score}`, but the page called `.join(", ")` on them as if they were strings.
08/11/2026	Agrata	**Fix:** the page now maps each role to its label and score, removes roles scoring 0%, and shows the top three. Strings are still handled for older saved profiles. Result for the data analyst sample CV: "Data Analyst (78%), Software Developer (32%), Cybersecurity Analyst (12%)".
08/11/2026	Agrata	**Investigated, not a bug:** "5 years of experience" for a CV described as a "recent graduate". `estimate_years()` merges overlapping date ranges, and the sample lists jobs from 2021 to Present, which is 5 years in 2026. The sample CV text is inconsistent, not the calculation.

Frontend UI, Branding and Theme Subsystem

The app now shows the CareerPilot logo in the sidebar, on the login page and as the browser-tab icon, and has a site-wide Light / Dark / Auto theme switch.

Files used

Files used
`frontend/public/logo.svg`
`frontend/app/icon.svg`
`frontend/app/globals.css`
`frontend/app/layout.tsx`
`frontend/app/dashboard/layout.tsx`
`frontend/app/login/page.tsx`
`frontend/components/ThemeToggle.tsx`
`frontend/components/themeScript.ts`

Date	Team member	Notes
08/11/2026	Agrata	**Logo:** the logo (navy rounded square with a white and light-blue paper plane, 120 × 120 SVG viewBox) replaced the plain teal placeholder square (`<i />`) in the sidebar and login page. Copied to `public/logo.svg` for display and `app/icon.svg`, which Next.js serves automatically as the favicon. Rendered a preview on both backgrounds: the navy square blended into the dark sidebar, so a 1 px translucent white outline was added there.
08/11/2026	Agrata	**Theme toggle problem:** dark mode only followed the operating system (`@media (prefers-color-scheme: dark)`), and the Theme option the team found in the Next.js developer toolbar changes only that toolbar, not the site.
08/11/2026	Agrata	**Theme implementation:** colours were already CSS variables, so a `data-theme` attribute on `<html>` now overrides them: `[data-theme="dark"]` forces dark, `:root:not([data-theme="light"])` inside the media query keeps OS-following when no choice is set. `color-scheme` is set so form controls match. A `ThemeToggle` client component (three buttons with `aria-pressed`) saves the choice to `localStorage`.
08/11/2026	Agrata	**Flash prevention:** a small inline script in the root layout's `<head>` applies the saved theme before first paint. It was first imported from the client component, which fails because a server component cannot import a plain value from a `"use client"` module; it was moved to `themeScript.ts`. `suppressHydrationWarning` was added

Date	Team member	Notes
		to `<html>`.
08/11/2026	Agrata	**Testing:** frontend type-check and production build passed (15 routes). A Playwright browser test confirmed that clicking Dark changes the page background to `rgb(14, 21, 32)`, the choice persists after reload, and Auto returns to the light system default.

Development Environment, Testing and Version Control

The app runs locally on Windows as two servers (FastAPI on port 8000, Next.js on port 3000), and the backend suite grew from 26 to 31 passing tests over this period.

Files used
`backend/.venv/` (local only, git-ignored)
`backend/.env`, `frontend/.env.local` (local only, git-ignored)
`backend/requirements.txt`, `frontend/package.json`
`.gitignore`, `.github/workflows/ci.yml`
`README.md`

Date	Team member	Notes
05/11/2026	Agrata	**Local hosting on Windows:** the README commands were written for macOS/Linux (`source .venv/bin/activate`). Ran the Windows equivalents in PowerShell: `py -m venv .venv`, `.venv\Scripts\Activate.ps1`, `pip install -r requirements.txt`, `copy .env.example .env`, `uvicorn app.main:app --reload`; then `npm install` and `npm run dev` for the frontend. First attempt failed because the commands were run in the project root instead of `backend`. The `{"detail":"Not Found"}` page at `127.0.0.1:8000` was confirmed as expected: the API has no root route, and `/docs` serves the interactive API documentation.
05/11/2026	Agrata	**LAN access documented:** start Uvicorn with `--host 0.0.0.0` and Next.js with `-H 0.0.0.0`, point `frontend/.env.local` at the PC's IP and add it to `CORS_ORIGINS`, so teammates on the same network can test.
07/11/2026	Agrata	**Virtual environments:** documented why each project uses its own `.venv` (isolated package versions, a reproducible

Date	Team member	Notes
		setup from `requirements.txt`, a clean global Python). VS Code's "Python: Select Interpreter" is recommended to be set to `backend\.venv` so new terminals activate it automatically.
07/11/2026	Agrata	**Git workflow:** confirmed `.env`, `.venv` and `careerpilot.db` are git-ignored (greyed out in VS Code) before committing. Documented a workflow of feature branches and pull requests (e.g. `feature/learning-links-gemini`) so the CI workflow runs the tests before merging to `main`.
08/11/2026	Agrata	**Clean-up:** a duplicate `favicon.svg` and an accidental `Contacts` folder (containing only a Windows `desktop.ini`) were found in the repository root and flagged for deletion before commit.

Date	Backend tests passing	Added
08/11/2026	31	3 Adzuna live-jobs tests
05/11/2026	28	2 learning-link tests
Before 05/11/2026	26	Original MVP suite (auth, upload validation, privacy between users, matching, reports)

Design and Style Constants

All colours are CSS custom properties in `frontend/app/globals.css`, so one change applies across every page and both themes.

Token	Light	Dark	Used for
`--bg`	#f5f6fa	#0e1520	Page background
`--card`	#ffffff	#162130	Cards and panels
`--ink`	#172033	#e8edf4	Body text
`--muted`	#5d6b82	#9aa8bd	Secondary text
`--line`	#e2e6ee	#26364a	Borders and dividers
`--brand`	#0f766e	#2dd4bf	Buttons, links, active states
`--nav`	#12202f	#0a111a	Sidebar background
`--good`	#067647	#4ade80	Highly qualified, success

Token	Light	Dark	Used for
`--warn`	#b54708	#fbbf24	Qualified with gaps, warnings
`--bad`	#b42318	#f87171	Development required, errors
Logo navy	#12305B	#12305B	Logo square
Logo accent	#FFFFFF, #BFD0EA	same	Paper plane

Typography: system font stack (`system-ui, -apple-system, "Segoe UI", Roboto, sans-serif`), 15 px body at 1.55 line height. Headings: h1 24 px, h2 16 px, h3 14 px, weight 650.

Shape: card corner radius 12 px; buttons 9 px; logo 28 × 28 px with an 8 px radius.

Match bands: 80 and above = Highly qualified (green), 60–79 = Qualified with gaps (amber), below 60 = Development required (red).

Database Schema

The database has 11 tables defined with SQLAlchemy in `backend/app/models.py`. On 08/11/2026, `jobs` gained `url` and `external_id` and the `job_query_cache` table was added for the Adzuna integration.

Table	Key columns	Purpose
`users`	id, email (unique), name, password_hash, created_at	Accounts; passwords stored as PBKDF2 hashes
`cv_documents`	id, user_id → users, filename, mime_type, size_bytes, text, is_active	Uploaded CVs and extracted text; only one active per user
`profiles`	user_id (PK) → users, summary, years_experience, education, experience, inferred_roles (JSON), target_roles, location, min_salary, experience_level, goals	Output of the Profile agent plus job preferences
`skills`	id, user_id → users, slug, name, category	Skills extracted from the CV (technical, tool, soft, certification)
`jobs`	id, title, company, location, role_category, level, min_years, salary_min, salary_max, description, skills (JSON), embedding (JSON), source, **url**, **external_id**	Seed and live postings; `source` = `seed-dataset` or `adzuna`
`job_query_cache`	id, key (unique: country|role|location), fetched_at, result_count	When each live search last ran; drives the 12-hour cache
`applications`	id, user_id → users, job_id → jobs, status, notes, updated_at; unique (user_id,	Application tracker

Table	Key columns	Purpose
	job_id)	
`reports`	id, user_id → users, payload (JSON), created_at	Saved readiness reports
`interview_sessions`	id, user_id → users, role, questions, answers, created_at	Interview practice history
`agent_runs`	id, user_id, agent, status, duration_ms, llm_source, note, created_at	LLMOps log shown on the Agent monitor page; `llm_source` = template, llm or template-fallback
`audit_logs`	id, user_id, action, detail, created_at	Security-relevant events (e.g. MATCH_CALCULATION)

Existing databases are upgraded at startup by `add_missing_columns()` in `main.py`, because `create_all()` creates new tables but never alters existing ones.

Learning Outcomes Evidence

Each unit learning outcome is backed by specific diary entries above.

Learning outcome	Evidence in this diary
Rationalise and propose diverse solutions for current and anticipated challenges	Compared four LLM options (Ollama, Groq, Gemini, OpenAI/Claude) and chose Gemini for remote team use. Compared six job APIs on coverage, quota, key access and terms, and chose Adzuna with JSearch as a later option. Anticipated quota limits (12-hour cache), API outages (seed fallback), schema changes on existing databases (startup migration) and stale course links (link checker).
Integrate the proposed solutions using the team's chosen technologies	Gemini connected through the existing OpenAI-compatible client and `.env` configuration. Adzuna added as a new `JobSource` in the FastAPI backend without changing the matcher, skill gap, learning or report agents. Curated learning catalogue, logo and theme toggle integrated into the Next.js frontend.
Experiment to verify the feasibility of proposed solutions	`check_llm`, `check_jobs` and `check_links` scripts test each external service against real endpoints. Mock-transport tests prove fallback behaviour; the backend suite grew from 26 to 31 passing tests. Migration tested on an old-schema database; the theme toggle tested in a scripted Playwright

Learning outcome	Evidence in this diary
	browser run; frontend production build verified.

Meeting Records

No meetings have been recorded yet; copy this template for each team or supervisor meeting.

Meeting 1

Field	Details
Project title	CareerPilot
Facilitator	
Team members	Agrata Shrestha, Ujjwol Adhikari
Date	
Time	
Location	
Meeting objective(s)	
Present	
Absent	

Agenda item	Discussion and outcome	Owner
Topic 1		
Topic 2		
Topic 3		

Next meeting	Details
Date	
Time	
Location	
Objective(s)	